

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

Marks

Q. 1 Solve Any Two of the following.

- A) Write short notes on: - 6
- a. Rock mass
 - b. Primary objectives of studying rock mechanics
 - c. How is rock mechanics used in the construction of tunnels?
- B) State and explain the principles and scope of rock mechanics. 6
- C) Discuss the major milestones in the development of rock mechanics and their impact on engineering practices. 6

Q.2 Solve Any Two of the following.

- A) Write a short note on: - 6
- a. What is flexural strength?
 - b. Tensile strength of a rock
 - c. What role does shear strength play in slope stability?
- B) Explain the procedures and equipment used in core drilling for rock sampling. 6
- C) Describe the Mohr-Coulomb failure criterion and its application in rock mechanics. 6

Q. 3 Solve Any Two of the following.

- A) Write a short note on: - 6
- a. Factors considered in RSR system
 - b. Why is RQD important in rock mechanics?
 - c. Name three basic types of rocks.
- B) Explain the procedure for determining the Rock Quality Designation (RQD) of a rock core sample. 6
- C) Describe the main parameters considered in rock mass classification systems. 6

Q.4 Solve Any Two of the following.

A) Write a short note on: -

- a. What factors influence the stability of rock slopes?
- b. Why is monitoring important for rock slope stability?
- c. How does planar failure occur?

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B) Describe the process and benefits of using grouting to improve rock properties.

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C) Discuss the various mechanical methods used to enhance the strength and stability of rock masses.

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Q. 5 Solve Any Two of the following.

A) Write a short note on: -

- a. Various types of deformations in tunnels.
- b. Type of rock support used in tunnels.
- c. Method used to measure rock stresses around a tunnel.

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B) Describe the methods used to assess the suitability of a rock mass for foundation construction.

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C) Discuss the various types of tunnel linings and their functions.

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***** End *****